

Safety Guidelines

ForgeCore Industries — Environment, Health & Safety

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1. Scope

These guidelines cover safe handling, storage, and use of ForgeCore crucible products. Compliance with these guidelines is mandatory for all personnel handling our products. Site-specific safety plans must incorporate these requirements and may add additional controls based on local hazard assessments.

2. Personal Protective Equipment (PPE)

Required PPE for crucible handling:

- Heat-resistant gloves rated to 250°C minimum (leather or aluminised)
- Safety glasses with side shields (ANSI Z87.1)
- Face shield for pour operations (additional to safety glasses)
- Long-sleeved flame-resistant clothing (Nomex or equivalent)
- Safety footwear with metatarsal protection
- Respiratory protection (N95 minimum) when handling dry powder materials

3. Material-Specific Hazards

Product	Primary Hazard	Control
FC-45 (SiC)	Dust inhalation during machining	Wet cutting, local exhaust, P100 respirator
FC-50 (Graphite)	Dust ignition above 500°C in air	Inert atmosphere during pre-heat
FC-72 (Clay-graphite)	Silica dust (cristobalite risk)	Wet handling, medical surveillance
FC-90 (Zircon)	Generally inert, dust nuisance	Standard dust controls

4. Storage Requirements

Crucibles must be stored in a dry, covered area protected from weather. Storage temperature should remain between 5-35°C with relative humidity below 70%. Crucibles must be stored upright on padded surfaces to prevent rim damage. Never stack crucibles inside each other. Maximum stack height: 3 crates.

5. Hot Handling Procedures

Crucibles removed from furnaces at operating temperature present severe burn and thermal shock risks. Use only ForgeCore-approved tongs and lift fixtures rated for the crucible weight plus a 50% safety margin. Allow hot crucibles to cool on a refractory surface for a minimum of 4 hours before any further handling.

6. Emergency Response

Burns:

Immediate cooling with clean running water for 20 minutes minimum

Do not apply ice directly; do not break blisters

Seek medical attention for any burn larger than a 50-paisa coin or on hands/face/joints

Inhalation of dust:

Move affected person to fresh air immediately

Administer oxygen if breathing is difficult

Seek medical attention if symptoms persist beyond 30 minutes

Crucible failure during pour:

Evacuate area within 5 metres radius

Allow molten metal to solidify before cleanup (minimum 2 hours)

Do not use water to extinguish molten metal — risk of steam explosion

Use dry sand or Class D fire extinguisher for metal fires

7. Training Requirements

All personnel operating crucible furnaces must complete ForgeCore's 2-day Crucible Safety and Operations course (CSO-101) before independent operation. Refresher training is required every 24 months. Records maintained in the ForgeCore Training Management System.